

Routing

A Single Page Application never actually reloads the browser — Angular's **Router** lets one page swap components based on the URL, giving you real navigation, bookmarkable links, and a working back button.

1 · ROUTES

Mapping URLs to components

```
export const routes: Routes = [  
  { path: 'dashboard', component: DashboardComponent },  
  { path: 'products', component: ProductsComponent },  
  { path: 'login', component: LoginComponent },  
];
```

A route is simply a rule: "when the URL looks like this, show that component." Angular checks the list top to bottom and renders the first match.

2 · ROUTER & ROUTER-OUTLET

Where the matching component appears

```
<nav>...</nav>  
<router-outlet></router-outlet>
```

URL changes

/products

**Router matches**

finds the route

**router-outlet**

swaps content

Think of `<router-outlet>` as a window frame that never moves — the Router just changes which picture is showing through it. The navbar, footer, and sidebar stay exactly where they are.

3 · TWO WAYS TO NAVIGATE

routerLink vs. the Router service

Declarative — routerLink

```
<a routerLink="/products">  
  Products  
</a>
```

Best for menus, nav bars, plain links.

Programmatic — Router service

```
constructor(private router: Router) {}  
  
goToLogin() {  
  this.router.navigate(['/login']);  
}
```

Best after logic runs first — e.g. redirect after a form submits.

4 · ROUTE PARAMETERS

Passing data through the URL with `:id`

```
{ path: 'products/:id', component: ProductDetailComponent }

constructor(private route: ActivatedRoute) {}
ngOnInit() {
  const id = this.route.snapshot.paramMap.get('id');
}
```

The colon marks a dynamic segment — it can match `/products/12`, `/products/45`, any id at all. One route definition, endless product pages. `ActivatedRoute` is the service Angular injects so a component can read exactly which URL activated it.

5 · A REAL APP'S ROUTES

A tiny routes array powers a whole app

PATH	COMPONENT
<code>/dashboard</code>	DashboardComponent
<code>/products</code>	ProductsComponent
<code>/products/:id</code>	ProductDetailComponent (nested)
<code>/login</code>	LoginComponent
<code>/profile</code>	ProfileComponent